MILITARY SPECIFICATION

CORES: AIRCRAFT HIGH-PRESSURE AIR VALVE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for one type of high pressure air valve core.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

QQ-N-290	-	Nickel Plating, Electrodeposited
QQ-B-654	-	Brazing Alloy, Silver
UU-P-553	-	Paper, Wrapping Tissue
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-T-60	-	Tape, Packaging, Waterproof

Military

MIL-P-116	-	Preservation, Methods of
MIL-H-5606	-	Hydraulic Fluid, Petroleum Base, Aircraft and Ordnance

STANDARDS

Federal

FED-STD-601	-	Rubber; Sampling and Testing

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-130	-	Identification Marking of U.S. Military Property
MIL-STD-143	-	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-794	-	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-831	-	Test Reports, Preparation of
MIL-STD-889	-	Dissimilar Metals
MS 24690	-	Packing, Preformed and High Pressure Air Valve

Air Force-Navy Aeronautical

AN809	-	Core - High Pressure Air Valve
AN6287	-	Valve - Air High Pressure 3/4 Hex
AND10071	-	Boss and Installations - Air Connection

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.2 Order of Precedence. In the event of a conflict between the text of this specification and the references cited herin, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Qualification. The valve core furnished under this specification shall be a product which is qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3, 4.3.1, and 6.3).

3.2 First article. When specified (see 6.2), the contractor shall furnish sample unit(s) for first article inspection and approval (see 4.4).

3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.4 Materials. Materials shall conform to applicable specifications and shall be as specified herein and on applicable drawings. Materials which are not covered by specifications, or which are not specifically described herein, shall be of the best quality, of the lightest practicable weight, and suitable for the purpose intended.

3.4.1 Metal parts. All metal parts shall be of a corrosion-resistant material or treated in a manner to render them adequately resistant to corrosion.

3.4.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

3.4.2 Nonmetallic materials. Elastomer components, except silicone, shall be not more than 12 months old from the date of manufacture to the date of delivery to any Government Service or to any airframe or accessory manufacturer.

3.5 Standard parts. MS and AN Standard parts shall be used where they suit the purpose.

3.6 Design and construction. The valve core shall be designed and constructed in accordance with AN 809. The valve core shall fit the valve body conforming to AN 6287.

3.6.1 Valve and plug seats. The valve and plug seats shall be constructed of an elastomer that is resistant to hydraulic fluid conforming to MIL-H-5606. The elastomer shall be homogeneous to insure consistent swell, strength and hardness characteristics.

3.6.2 Brazing and soldering. Metal parts of the valve core assembly shall be joined by copper furnace brazing or soldering in accordance with QQ-B-654. Crimping or swaging shall not be used to join metal parts in the valve core assembly, where expansion or contraction would allow leakage of air through the valve.

3.6.3 Finish. Exposed metal surfaces shall be protected by nickel plating conforming to QQ-N-290.

3.7 Performance.

3.7.1 Physical properties. The tensile strength, elongation, and hardness of the elastomers used for the valve and plug seats shall not change more than is indicated in Table I, when subjected to accelerated aging.

TABLE I
Allowable Change in Physical Properties of Valve Seat Material Due to Aging

Property	Change	
	Increase	Decrease
Tensile strength	-	20 percent max
Elongation	-	30 percent max
Hardness (Shore durometer), points	10	10

3.7.2 Swelling. The elastomer used for the valve and plug seats shall withstand immersion in hydraulic fluid conforming to MIL-H-5606 without increasing in volume by more than 12 percent. The elastomer shall not shrink.

3.7.3 Leakage. The valve cores shall not leak when tested in accordance with paragraph 4.6.4.

3.7.4 Proof pressure. The valve cores shall be subjected to proof pressure of 6,000 pounds per square inch in accordance with paragraph 4.6.5.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

3.7.5 Burst pressure. The valve cores shall not fail when subjected to a burst pressure of 12,000 pounds per square inch in accordance with paragraph 4.

3.7.6 Endurance. The valve cores shall not leak or malfunction when subjected to the endurance test in accordance with paragraph 4.

3.8 Marking. In addition to the marking specified on Drawing AN809, each valve core shall be marked with the manufacturer's name, symbol, or trademark in accordance with MIL-STD-130.

3.8.1 The brass portion of the swivel plug on each valve core shall be copper-plated to facilitate its identification as a high-pressure core conforming to this specification. The copper plate shall be the result of 1 hour plating at 6 volts in a plating barrel, followed by a cold water rinse. The plating shall be protected by a coating of clear lacquer compound or equal parts lacquer and thinner.

3.9 Workmanship. The finished valve cores shall be free from defects. Workmanship and finish shall conform to the highest grade commercial practice in the manufacture of this class of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The examination and testing of the valve cores shall be classified as follows:

(a) Qualification inspection. Qualification inspection consists of examinations and tests performed on samples submitted for approval as qualified products.

(b) First article inspection. First article inspection consists of examinations and tests performed on samples which are representative of the production item after award of a contract to determine that the production item meets the requirements of this specification.

(c) Quality conformance inspection. Quality conformance inspection consists of examinations and tests performed on individual products or lots to determine conformance of the products or lot with the requirements set forth in this specification.

4.3 Qualification inspection. Qualification inspection shall consist of all the examinations and tests of this specification.

4.3.1 Qualification samples. Qualification samples shall consist of the following: A minimum of six valve cores, representing the manufacturer's

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

production item and conforming to the detail drawings presented with his request for authorization of Qualification tests; a flat sheet approximately 1/8 by 6 by 12 inches in size of the same compounding and cure as each seat and plug material to be used in the construction of the valve cores. Samples shall be identified as required and forwarded to the agency or agencies designated in the Letter of Authorization.

4.3.2 Inspection report. Three copies of the qualification inspection report shall be forwarded with the samples specified in 4.3.1. One copy shall be reproducible. The report shall be prepared in accordance with MIL-STD-831 and shall include the results of all examiniations and tests, including a detailed statement indicating compliance or extent of noncompliance with each requirement of this specification, referring specifically to the applicable paragraph number.

4.4 First article inspection. Unless otherwise specified, samples specified in 4.4.1 shall be submitted for the performance of examinations and tests deemed necessary by the procuring activity [see 6.2 (c)].

4.4.1 First article samples. First article samples shall consist of three valve cores. The samples shall be representative of the construction, components, materials, and workmanship used during production.

4.4.1.1 Submitting samples. Unless otherwise specified, as soon as practicable after the award of the contract or order, samples specified in 4.4.1 shall be forwarded as specified in the contract or order. The samples shall be plainly identified by securely attached durable tags marked with the following information:

> Sample submitted by (name) (date) for first article inspection in accordance with MIL-C-5499 under contract No. _____.

When a contractor is in continuous production of these valve cores from contract to contract, submission of further samples on the new contract may be waived at the discretion of the procuring activity. Approval of the first article samples or the waiving of the first article inspection does not preclude the requirements for performing the quality conformance inspections.

4.4.1.2 Disposition of samples. Upon completion of the first article inspections, all the applicable inspection reports and, when applicable, recommendations and comments pertinent for use in monitoring production will be forwarded to the Government quality control representative. The valve cores will be consumed or destroyed in the first article inspection and shall not be considered as part of the quantity to be delivered under contract.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the tests specified herein. When inspection is conducted at the contractor's plant, all inspection and testing shall be under the supervision of the Government Inspector. Contractors not having laboratory testing facilities satisfactory to the Government shall engage the services of a commercial testing laboratory acceptable to the Inspector. The contractor shall furnish test reports, in duplicate, showing quantitative results for all tests required by this specification, and signed by an authorized representative of

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

the contractor or laboratory, as applicable. Acceptance or approval of material during course of manufacture shall in no case be construed as a guaranty of the acceptance of the finished product.

4.5.1 Sampling. A flat sheet approximately 1/8 by 6 by 12 inches shall be obtained from each batch of plug and seat material used in the construction of valve cores. This sample shall be selected and cured under the supervision of the Inspector. In addition, six representative sample valve cores, selected from a lot by the Inspector, may be required if the Inspector deems it desirable to conduct any further tests specified herein. A lot shall consist of all high-pressure air valve cores which are constructed from the same plug and seat material and submitted for inspection under the same contract or order.

4.5.1.1 Samples subjected to destructive tests shall be in addition to the quantity specified in the contract or order, and shall be furnished without additional cost to the Government.

4.5.1.2 Rejection and retest. Failure of any sample of valve cores to conform to any one of the requirements of this specification shall be cause for the rejection of the lot, order, or design represented, as applicable. Valve cores which have been rejected may be changed to correct the defects found in the original and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the Government. Where failure of a sample or specimen is definitely ascribed to faulty material, the entire lot, order, or design represented, as applicable, shall be rejected.

4.6 Tests. The tests in paragraph 4.6.4 through 4.6.7 shall be conducted on six or more valve cores that are installed in a manifold or air tight container in accordance with Figure 1. These tests shall be conducted in the given sequences using the same cores in all tests.

4.6.1 Examination of product. Each valve core shall be carefully examined to determine conformance with Drawing AN809, and the requirements for workmanship and design as specified herein.

4.6.2 Physical properties. The tensile strength, elongation, and hardness of the plug and seat material shall be determined in accordance with method 4111, 4121, & 3021 of FED-STD-601. Samples of the material shall then be subjected to accelerated aging by the oven method in accordance with method 7001 and 7221 of FED-STD-601, for a period of 7 days. The tensile strength, elongation, and hardness shall be determined on the aged specimens, and the percentage difference between the aged and unaged values recorded.

4.6.3 Swelling. Swelling of plug and seat material shall be determined in accordance with the method 6001 & 12311 of FED-STD-601, except that the specimens shall be 1/8 by 1 by 2 inch blocks obtained from the flat sheets, and the period of immersion shall be 7 days. The solvent shall be hydraulic fluid conforming to Specification MIL-H-5606, maintained at a temperature of 70° ± 1°C (158 ± 2°F).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

4.6.4 Leakage. There shall be no evidence of leakage as a result of the following tests.

4.6.4.1 Impulse. The valve cores shall be subjected to pressure impulse between pressures of 300 to 1500 psi, under the conditions and in the sequence shown in table II. The presure shall not be allowed to fall below 200 pounds per square inch at any time during the test. No valve caps shall be used to cover the cores. An air tight rubber or synthetic bladder shall be installed over each valve body in a collapsed condition in such a manner as to collect separately all air leakage past each valve core. These bladders shall not be inflated in any proportion whatsoever during the full period of this test.

TABLE II
Impulse Conditions
(Air in Contact with Valve Cores)

Operation	Pressures	Temperature	Time	Total Cycles at This Condition
Holding Pressure Under Heat	200 psi min	71°C (160°F) min	6 hrs (min)	
Cycling Under Heat	300 ±100 psi to 1500 +500 -0 psi	71°C (160°F) min	35 cycles per min (max)	10,000
Holding Pressure Under Cold	200 psi min	Not warmer than -54°C (-65°F)	72 hrs (min)	
Cycling Under Cold	300 ±100 psi to 1500 +500 -0 psi	Not warmer than -54°C (-65°F)	35 cycles per min (max)	5,000
Normal Cycling	300 ±100 psi to 1500 +500 -0 psi	Room temp approx 21°C (70°F)	35 cycles per min (max)	35,000

4.6.4.2 Temperature and pressure variations. The valve cores shall be subjected to pressure and temperature cycling in accordance with table III. These cycles and their portions shall be conducted consecutively but may be interrupted providing no appreciable drop in pressure occurs during the intervals between portions of the cycles. The manifold or container with the installed valve cores shall be connected to a pressure accumulator containing a minimum of 50 cu in. of air, for each valve on test. The manifold or container shall be placed in a temperature control box capable of maintaining a free air temperature of from -57° to +71°C (-70 to +160°F). Inflation of the bladders covering the cores in any degree whatsoever during the total period of this test shall indicate leakage of the valve cores and shall be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

TABLE III
Pressure and Temperature Conditions
(of Air in Contact with Valve Cores)

Cycle	Pressure psi	Conditions of Each Portion of Cycle Temperature deg. cent. (deg. fahr.)	Duration of each Portion of Cycle hr. ±1/2 hr	Number of Complete Cycles min.
A	70	71 (160) min	6	4
	70	21 ±8 (70 ±15)	18	
B	1500	71 (160) min	6	4
	1500	21 ±8 (70 ±15)	18	
C	70	-54 (-65) max	30	4
	70	21 ±8 (70 ±15)	18	
D	1500	-54 (-65) max	30	4
	1500	21 ±8 (70 ±15)	18	

4.6.5 Proof pressure. The sample valve cores shall be subjected to 6000 pounds per square inch pressure for a period of at least 5 minutes. No valve cap shall cover the core. The rate of applying pressure shall not exceed 25,000 pounds per square inch per minute. There shall be no evidence of leakage or damage to the valve seat material under this pressure. Inflation of the bladders covering the cores in any degree whatsoever during the test shall indicate leakage of the valve cores.

4.6.6 Endurance. There shall be no evidence of leakage or malfunctioning as a result of the following tests.

4.6.6.1 Inlet air. High-pressure air shall be applied to the valve cores in interrupted flow according to table IV. The pressure of the inlet air through the valve cores shall be from 85 pounds per square inch to a maximum pressure great enough to overcome valve core spring differential and exhaust an equivalent volume of air through a relief valve in the manifold or container. The relief valve shall be set to a minimum pressure of 600 pounds per square inch.

4.6.6.2 Discharging air. The direction of the air through the valve shall then be reversed. A continuous supply of air at conditions conforming to table IV shall be available to the manifold or container. The valve stem shall be depressed by mechanical means sufficient to allow free exit of air on each depression.

4.6.7 Burst pressure. The sample valve cores shall be subjected to a burst presure of 12,000 pounds per square inch for a period of at least 5 minutes. No valve cap shall cover the core. The rate of applying pressure shall not exceed 25,000 pounds per square inch per minute. There shall be no failure of any portion of the valve.

Source: https://assist.dla.mil -- Downloaded: 20261O-17T04:54Z
Check the source to verify that this is the currentversion before use.

TABLE IV
Endurance Test Conditions

	Pressure psi	deg C (deg F)	Valve Openings per Minute (±10)	Cycles of Operation (min)
Admitting Air into Valve	85^{+15}_{-0} to 600 min	39 to 60 (102 to 140)	35	50,000
Discharging Air Out of Valve	600 ±50	21 ±6 (70 ±10)	25	1,000
Discharging Air Out of Valve	600 ±50	-54 (-65) max. Hold valves and air supply at this temp and req'd pressure for 36 hrs before discharging air	as req'd	10

5. PACKAGING

5.1 Preservation - packaging. Preservation-packaging shall be in accordance with MIL-STD-794, Level A or C as specified by the procuring activity.

5.1.1 Level A. Each core shall be preserved and packaged in accordance with Method 1A of MIL-P-116, without the application of a contact preservative. Cleaning shall be accomplished without immersion in solvent. The item shall be wrapped in tissue wrapping paper conforming to UU-P-553 and placed within the watervaporproof enclosure to accomplish this method and sealed.

5.1.2 Level C. Each unit shall be packaged in a manner that will prevent physical damage or deterioration during transit from the supply source to the first receiving activity for immediate use. The package shall be capable of precluding contamination that might impair the use or efficiency of the core. The contractor's commercial packaging may be utilized when it conforms to the criteria stated herein.

5.2 Packing. Unless otherwise specified by the procuring activity, the core shall be provided packing which will afford adequate protection against damage during shipment, handling, intermediate storage and worldwide redistribution.

5.2.1 Level A. Cores packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style RSC, grade V3c or V3s. As far as practical, exterior containers shall be uniform in shape and size, be of minimum cube and tare consistent with the protection required, and contain identical quantities. Closure and waterproofing shall be accomplished by sealing all seams, corners and

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60. Reinforcement requirements shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding.

5.2.2 Level B. Cores packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to style RSC, class domestic, minimum grade 200 of PPP-B-636. Containers shall be of uniform shape and size and contain identical quantities. The container closure shall be accomplished in accordance with Method II, 30.3.1.2 of the appendix to PPP-B-636.

5.2.3 Level C. The packaged cores shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations.

5.3 Marking. All unit and shipping containers shall be marked in accordance with MIL-STD-129. In addition, the following precautionary marking shall appear on the unit and shipping container:

CAUTION

OXYGEN COMPONENTS-
ALL OIL, GREASE, SHOP
RESIDUE, OR OTHER
CONTAMINATES HAVE
BEEN REMOVED.

Do not open until ready for use.

6. NOTES

6.1 Intended Use. The valve cores covered by this specification are intended for use in high-pressure, air-charged units such as shock absorber struts, hydraulic pressure accumulators, and surge cylinders.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification

(b) Applicable Government part number (AN 809-1)

(c) If first article inspections are to be performed and if so, instructions for forwarding the samples (see 4.4)

(d) Applicable levels of preservations, packaging, and packing (see 5.1)

(e) Data requirements (see 6.4)

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractor is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is OC-ALC/MMEDO, Tinker AFB OK 73145, and information pertaining to qualification of products maybe obtained from that activity.

6.4 Data. For the information of contractors and contracting officers, any of the data specified in applicable documents listed in section 2 of this specification or referenced lower-tier documents need not be prepared for the Government and should not be furnished to the Government unless specified in the contract or order. The data to be furnished should be listed on DD Form 1423 (Contractor Data Requirements List), which should be attached to and made a part of the contract or order.

6.5 Changes from previous issue . The margins of this specification are not marked to indicate where changes where made due to the extensiveness of the changes.

Custodians:
Air Force - 99
Army - AV
Navy - AS

Preparing Activity:
Air Force - 71

Project number:
1650-0378

☆U.S. GOVERNMENT PRINTING OFFICE: 1984 — 705-040/A4507

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

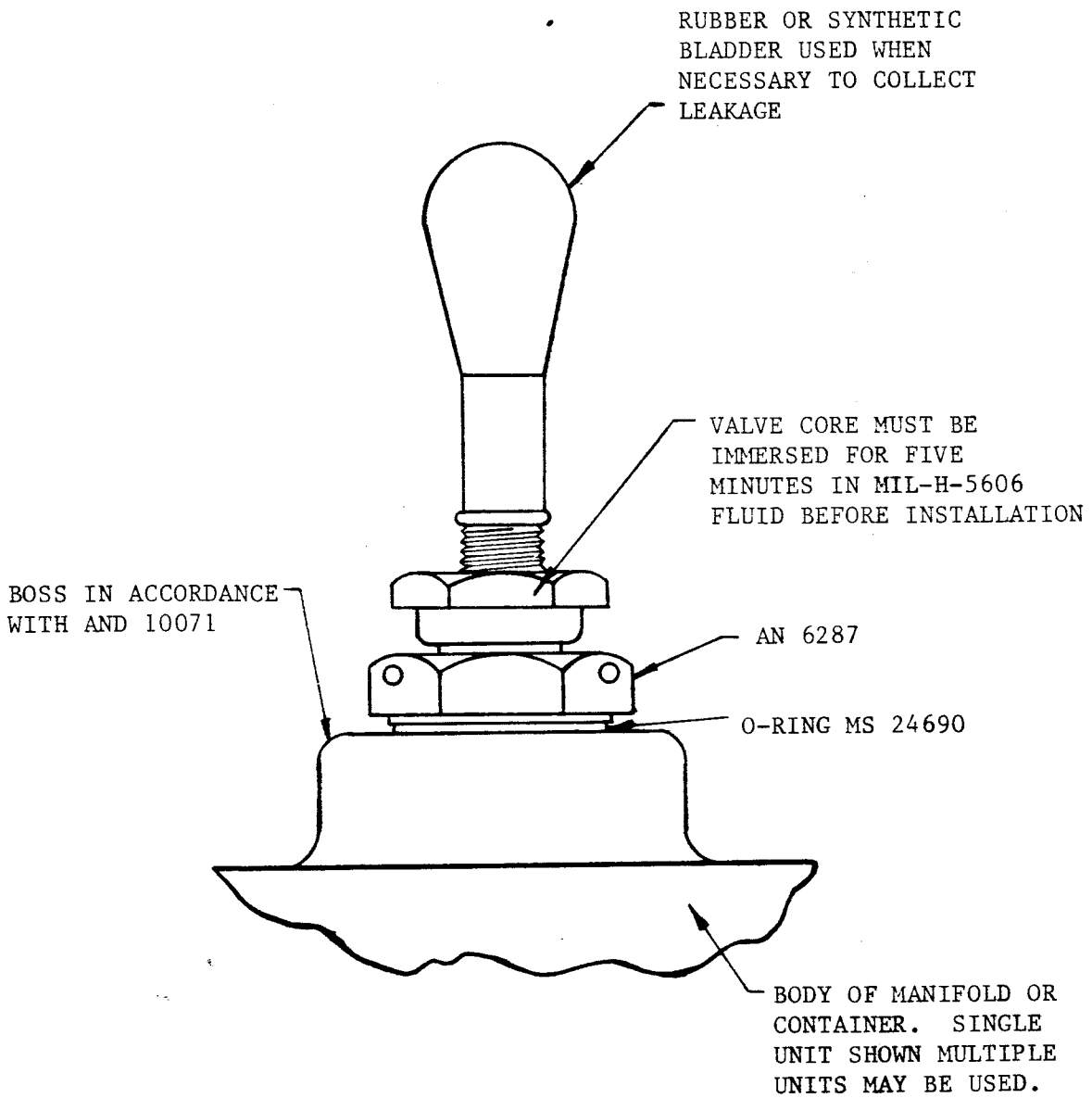


Figure 1. Typical valve core installation for performance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T04:54Z
Check the source to verify that this is the current version before use.